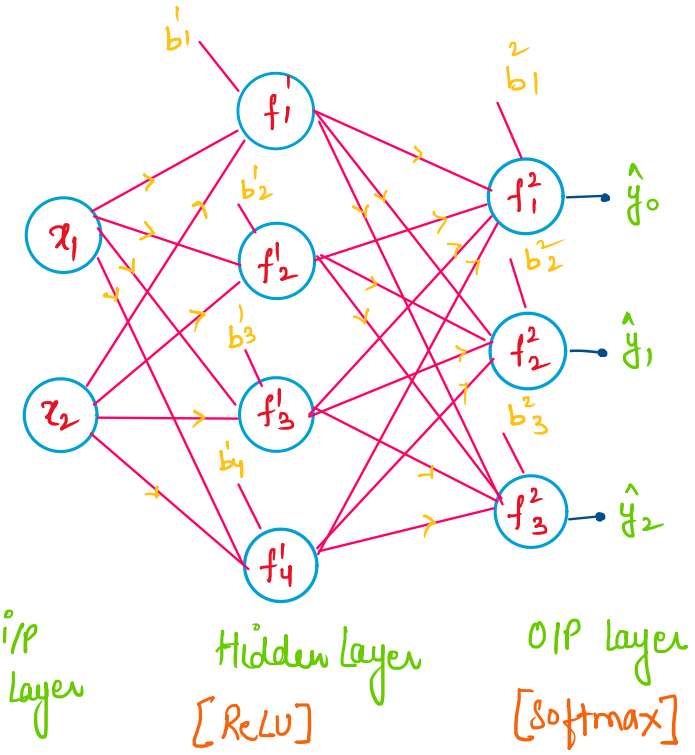
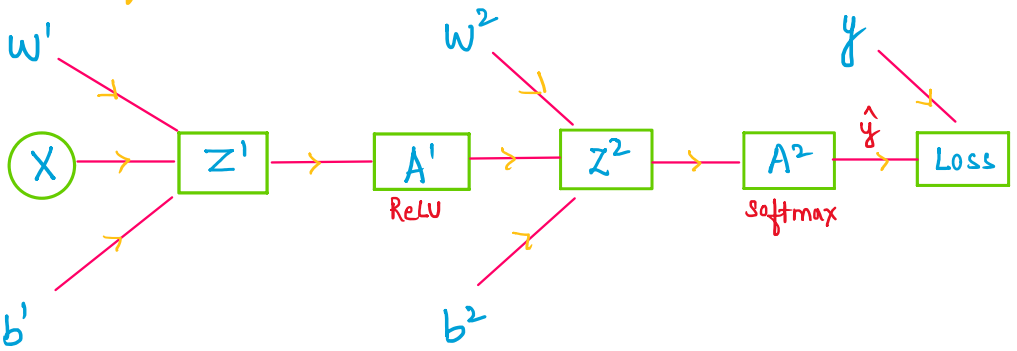
Neural Networks-4

★ N- Layer Backpropagation :-

N=2



Compressed form:-



X = input for Layer-1

W^1 & b^1 = parameters of Layer-1

A^1 = Activation func. of Layer1 & i/p of Layer-2

W^2 & b^2 = parameters of Layer-2

A^2 = Activation func. of Layer-2

y = Actual o/p

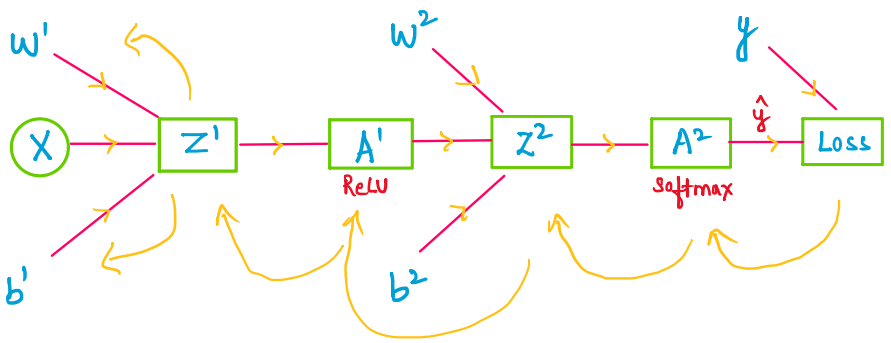
L = Loss

Z^1 = Real valued no. at Layer 1

Z^2 = Real valued no. at Layer 2

∵ we need to find the values of w^1, b^1, w^2 & b^2 in order to update these parameters

So,



Back propagation

$$\frac{\partial L}{\partial w^1} = \frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial Z^2} \cdot \frac{\partial Z^2}{\partial A^1} \cdot \frac{\partial A^1}{\partial Z^1} \cdot \frac{\partial Z^1}{\partial w^1}$$

So,

$$Z^1 = X \cdot W^1 + b^1$$

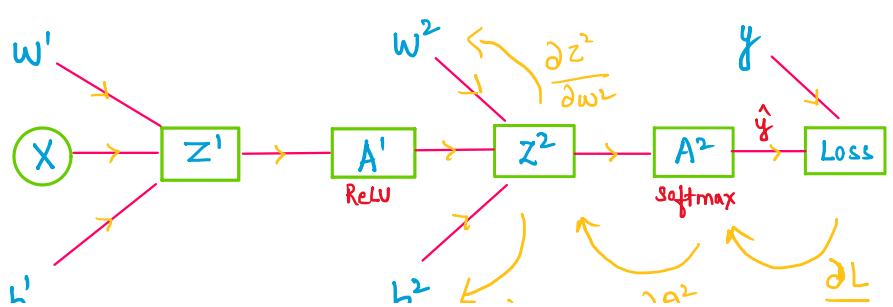
$$A^1 = \text{ReLU}(Z^1)$$

$$Z^2 = A^1 \cdot W^2 + b^2$$

$$A^2 = \text{softmax}(Z^2)$$

$$L = -\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{K} y_j \log(P_j)$$

Back Propagation from Layer 2 to 1 :-



b^1 $\qquad$ b^2 $\qquad$ $\frac{\partial z^2}{\partial b^2}$ $\qquad$ $\frac{\partial A^2}{\partial z^2}$ $\qquad$ $\frac{\partial L}{\partial A^2}$

Calculating for W^2 :-

$$\frac{\partial L}{\partial w^2} = \boxed{\frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial z^2}} \cdot \frac{\partial z^2}{\partial w^2} \Rightarrow \frac{\partial [A^1 \cdot w^2 + b^2]}{\partial w^2} = A^1$$

$$\Downarrow$$

$$P_k - I(i=k) = dz^2$$

So,

$$\frac{\partial L}{\partial w^2} = dz^2 \cdot A^1$$

or

$$\boxed{dW^2 = dz^2 \cdot A^1}$$

Calculating for b^2 :-

$$\frac{\partial L}{\partial b^2} = \underbrace{\frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial z^2}}_{dz} \cdot \frac{\partial z^2}{\partial b^2} \hookrightarrow \frac{\partial (A^1 \cdot w^2 + b^2)}{\partial b^2}$$

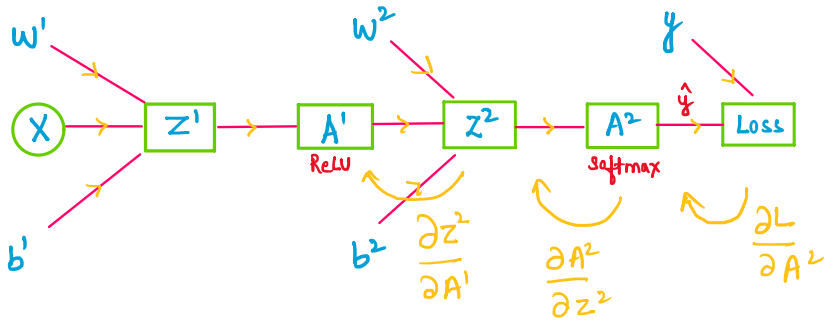
$$\frac{\partial L}{\partial b^2} = dz^2 \cdot 1$$

or

$$\boxed{db^2 = dz^2}$$

Calculating for A^1 :-

w^1 $\qquad$ w^2 $\qquad$ y



$$\frac{\partial L}{\partial A^1} = \boxed{\frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial z^2}} \cdot \frac{\partial z^2}{\partial A^1}$$

$$\Downarrow \quad dz^2 \qquad\qquad \hookrightarrow \quad \frac{\partial}{\partial A^1}\left(A^1 \cdot W^2 + b^2\right) = W^2$$

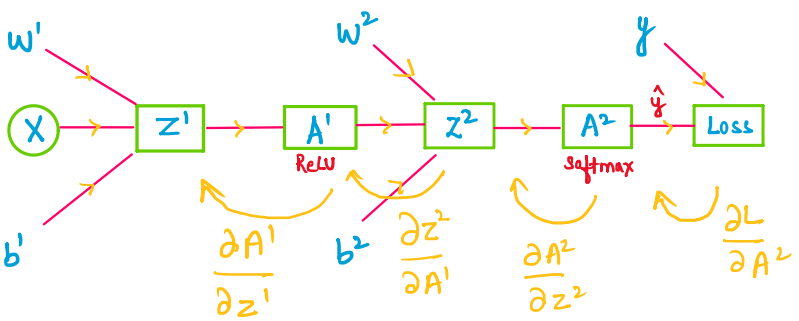
so,

$$\frac{\partial L}{\partial A^1} = dz^2 \cdot W^2$$

or

$$\boxed{dA^1 = dz^2 \cdot W^2}$$

Calculating for Z^1:-



$$\frac{\partial L}{\partial z^1} = \underbrace{\frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial A^1}}_{dA^1} \cdot \frac{\partial A^1}{\partial z^1}$$

$$A^1 = \mathrm{ReLU}(z^1) = \begin{cases} z, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

$$\frac{\partial A^1}{\partial z^1} = \mathrm{ReLU}'(z^1) = \begin{cases} 1, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

so, $dA^1 \cdot 1$

∂z^1 $\quad \begin{cases} 0 & , z \leq 0 \end{cases}$

so, $dA^1 \cdot 1$

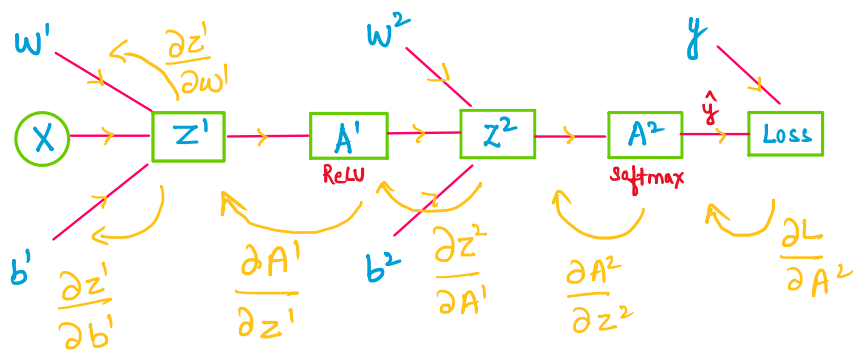
$dA^1 \cdot 0$

$$\therefore \frac{\partial L}{\partial z^1} = \begin{cases} dA^1 \times 1 & , z > 0 \\ dA^1 \times 0 & , z \leq 0 \end{cases}$$

or

$$\boxed{dz^1 = \begin{cases} dA^1 \times 1 & , z > 0 \\ dA^1 \times 0 & , z \leq 0 \end{cases}}$$

Calculating for W^1 and b^1 :-



$$\frac{\partial L}{\partial w^1} = \underbrace{\frac{\partial L}{\partial A^2} \cdot \frac{\partial A^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial A^1} \cdot \frac{\partial A^1}{\partial z^1}}_{dz^1} \cdot \frac{\partial z^1}{\partial w^1}$$

$$\Downarrow$$

$$\frac{\partial}{\partial w^1}\left[X \cdot W^1 + b^1\right]$$

$$\frac{\partial L}{\partial w^1} = dz^1 \cdot X$$

or

$$\boxed{dW^1 = dz^1 \cdot X}$$

Similarly,

Similarly,

$$db^1 = dz^1 \cdot 1$$

or $$\boxed{db^1 = dz^1}$$

from the above derivation, we come to know that -

① Calculating dz^2 gives :-

$$\underbrace{dW^2, db^2}_{\text{parameters of the } 2^{nd} \text{ Layer}}, dA^1$$

② Calculating dA^1 gives dz^1

③ Calculating dz^1 gives

$$\underbrace{dW^1, db^1}_{\text{parameters of the } 1^{st} \text{ layer}}$$

In order to not calculate the value of deeper derivatives again & again, we can calculate and store the derivatives of deeper layers. This is called as memoization.

- In case of linear regression problem, every thing will remain the same instead of the activation function in the O/P Layer.

	Classification	Regression
Activation →	softmax	Linear

Activation →	Softmax (because we want prob)	Linear (because we want a continuous o/p)